

Package ‘RAFM’

October 25, 2013

Type Package

Title Admixture F-model

Version 1.2

Date 2013-10-25

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Description Fits AFM by Metropolis-Hastings, calculates theta and FST out of AFM parameters

License GPL-2

LazyLoad yes

R topics documented:

RAFM-package	1
AFM	2
do.all	3
gen.fst	4
gen.theta	5
heat.theta	6
shrews	7
specimen	8
sticklebacks	9

Index	11
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RAFM-package	<i>Admixture F-model by Metropolis-Hastings</i>
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Description

This package involves codes needed for fitting the AFM from command line, conversion of AFM to coancestry and FST, and three example datasets. * * New feature in version 1.2: you can use function do.all to do all analyses! * *

Details

```

Package:    RA FM
Type:       Package
Version:    1.2
Date:       2013-10-24
License:    GPL-2
LazyLoad:   yes

```

In versions 1.2+, the primary mode to use this package is to run `do.all` with sufficient sample size.

If convergence problems occur, you should first increase the sample size, and then look into lower level, i.e. into AFM and `gen.theta`. If that does not help, you are free to play around with the source code (available at the software page, EGRU, University of Helsinki).

The code can be very slow for polymorphic data, such as shrews and sticklebacks in this package, so please be patient.

Author(s)

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References

Karhunen & Ovaskainen (2012): Estimating population-level coancestry coefficients by an admixture F-model. *Genetics* 192: 609-617.

Examples

```

data(specimen)
samp <- do.all(specimen, 100, 50, 2)

```

AFM

Function to fit the Admixture F-model (AFM) by MCMC

Description

The function takes in genotype matrix, and returns (1) posterior of lineage loadings κ and (2) posterior of drift parameters $\log(\alpha)$. Typically you will not need to use AFM.

Usage

```
AFM(dat, nMC, burnin, thinning, eps=7, priq=1, pria=c(1,2), prik=1)
```

Arguments

<code>dat</code>	data matrix, format as in dataset <code>specimen</code>
<code>nMC</code>	total number of MCMC iterations
<code>burnin</code>	no. of burnin iterations to be cast away, adjustment of proposal distributions is done during burnin
<code>thinning</code>	interval to sample from the MCMC, e.g. <code>thinning=10</code> samples every 10th iteration

eps	truncation threshold of the Dirichlet distributions. eps=7 truncates at 1e-07, eps=8 at 1e-08, etc.
priq	specifies prior for ancestral allele frequencies, Dirichlet(priq)
pria	specifies a prior for log(alpha): first component mean, second variance
prik	specifies a prior for kappa

Details

Convergence of the algorithm needs to be checked by the investigator (e.g. by doing posterior diagnostic tests or running a number of parallel chains.) As for the sample size, nmc=20000, burnin=10⁴, thinning=10 would typically be enough, but that depends on the case. The code can be fairly slow for highly polymorphic data such as microsatellites, so try short test runs. Results may require running the code overnight. There has to be at least two subpopulations in the data. (Insert a dummy row, if you want to estimate within-population theta for one subpopulation.)

The default priors should be fairly uninformative. priq=1 implies a uniform prior, higher values indicating more symmetric frequencies. pria=c(1,2) should be flexible enough for all biologically relevant scenarios. prik scales the diagonal-dominated Dirichlet prior of kappa, higher values implying more information.

Value

kapm	Posterior of lineage loadings kappa. Dimension is (population x lineage x mc iteration).
alpham	Posterior of lineage-specific drift parameters log(alpha). Dimension is (lineage x mc iteration). Note that this is posterior of LOG(alpha), thus negative values are ok.

Author(s)

Markku Karhunen, Uni. Helsinki

Examples

```
data(specimen)
samp <- AFM(specimen, 20, 10, 2) # few iterations!
postkap <- samp[[1]] # posterior of lineage loadings
hist(postkap[1,1,]) # histogram from population 1's loading to lineage 1
hist(postkap[1,2,]) # histogram from population 1's loading to lineage 2
```

do.all

Estimating coefficients of coancestry by admixture F-model

Description

FST values and coancestry matrix by push of a button.

Usage

```
do.all(dat, nMC, burnin, thinning, eps=7, priq=1, pria=c(1,2), prik=1)
```

Arguments

<code>dat</code>	data matrix, format as in dataset specimen
<code>nMC</code>	total number of MCMC iterations
<code>burnin</code>	no. of burnin iterations to be cast away, adjustment of proposal distributions is done during burnin
<code>thinning</code>	interval to sample from the MCMC, e.g. thinning=10 samples every 10th iteration
<code>eps</code>	truncation threshold of the Dirichlet distributions. <code>eps=7</code> truncates at $1e-07$, <code>eps=8</code> at $1e-08$, etc.
<code>priq</code>	specifies prior for ancestral allele frequencies, <code>Dirichlet(priq)</code>
<code>pria</code>	specifies a prior for $\log(\alpha)$: first component mean, second variance
<code>prik</code>	specifies a prior for κ

Details

This function performs all functionality of RAFM in one go, i.e. it's a wrapper of AFM and the subsequent parameter transformations `gen.theta` & `gen.fst`. The output is a list which contains samples from the posterior distributions of population-genetic parameters. The user needs to play around with AFM and/or different values of input parameters, if convergence problems occur.

Value

<code>theta</code>	Posterior of coancestry matrix θ . Dimension is (population x population x mc iteration).
<code>fst</code>	Posterior of Wright's fixation index. Each number is an mc iteration.
<code>kap</code>	Posterior of lineage loadings κ . Dimension is (population x lineage x mc iteration).
<code>alpha</code>	Posterior of drift parameters α . Dimension is (lineage x mc iteration).

Author(s)

Markku Karhunen, Uni. Helsinki

Examples

```
data(specimen)
samp <- do.all(specimen, 100, 50, 2)
```

`gen.fst`

Calculation of FST out of a coancestry matrix.

Description

A very simple function of the θ matrix. Typically you will not need to touch this.

Usage

```
gen.fst(theta)
```

Arguments

theta population-to-population coancestry matrix

Details

Weir-Cockerham definition of FST, known to be consistent with Wright and coalescent theory.

Value

Scalar number in [0,1].

Author(s)

Markku Karhunen, Uni. Helsinki

References

Cockerham CC & Weir BS (1987): Correlations, descent measures - drift with migration and mutation. PNAS 84: 8512-8514.

Examples

```
theta <- 0.15*diag(3)
gen.fst(theta)
```

gen.theta

Conversion of AFM parameters to coancestry matrix and FST.

Description

The AFM parameters involve the lineage loadings (kappa) and the drift parameters (alpha). Theta is the coancestry matrix, and FST is Wright's fixation index calculated out of theta. This function does the deterministic conversion of posteriors: (kappa, alpha) -> (theta, FST).

Usage

```
gen.theta(kapm, alpham)
```

Arguments

kapm posterior of the kappa matrix, given by function AFM. Dimension must be (population x lineage x mc iteration).

alpham posterior of the log(alpha) vector, given by the function AFM. Dimension must be (lineage x mc iteration).

Details

Note that input 2 is the posterior of LOG(alpha), i.e. it goes on a scale from minus to plus infinity.

Value

thetam	posterior of coancestry matrix, dimension (population x population x mc iteration)
fstm	posterior of FST

Author(s)

Markku Karhunen, Uni. Helsinki

Examples

```
data(specimen)
post_AFM <- AFM(specimen, 10, 4, 2) # a silly posterior with too few iterations
kap <- post_AFM[[1]]
logalpha <- post_AFM[[2]]
post <- gen.theta(kap, logalpha) # conversion
n <- sample(1:3, 1)
post[[1]][,n] # one sample from theta's posterior, should be psd
```

heat.theta

Illustrating neutral genetic differentiation

Description

This function illustrates the posterior of coancestry matrix theta as a heat map.

Usage

```
heat.theta(thetapost, mean=TRUE, verbose=TRUE)
```

Arguments

thetapost	posterior of theta
mean	mean=TRUE uses posterior means, mean=FALSE uses posterior medians
verbose	verbose=TRUE returns you summary statistics

Details

This very simple function calculates the posterior mean/median of the coancestry matrix theta, and uses R's standard heatmap to illustrate it. Do not blame me for the dendrogram or choice of colors. More sophisticated illustrations can be produced by manipulating the posterior manually.

Ugly averaging of off-diagonals takes care that the output matrices are positive definite, as they should for a sound theta.

Value

If verbose=TRUE, you will get a list that has three components: posterior mean/median, and 95 percent posterior credible interval of the theta matrix.

Examples

```
# Example commented away for obvious reasons
# data(specimen)
# samp <- do.all(specimen, 2000, 1000, 2) # <- obvious
# stats <- heat.theta(samp$theta, mean=FALSE, verbose=TRUE)
# names(stats)
# round(stats$median, 2) # and copy-paste to manuscript

# I do not claim responsibility for the dendrogram!
```

shrews

*Genotype data from Finnish common shrews (Sorex araneus)***Description**

149 shrews from 5 subpopulations genotyped for 20 microsatellite markers

Usage

```
data(shrews)
```

Format

A data frame with 149 observations on the following 21 variables.

```
subpop  subpopulation code
X9      allelic copy 1, locus X9
X9.1    copy 2, locus X9
X14     copy 1, locus X14
X14.1   ...
X33     ...
X33.1   ...
X45     ...
X45.1   ...
X62     ...
X62.1   ...
X67     ...
X67.1   ...
X68     ...
X68.1   ...
X69     ...
X69.1   ...
X92     ...
X92.1   ...
X97     ...
X97.1   ...
```

Details

Each row represents one individual. In this case, the genotypes are microsatellite fragment lengths. Missing data is coded as NA.

Source

Hanski I & Kuitunen J (1986): Shrews on small islands: epigenetic variation elucidates population stability. *Holarctic Ecology* 9: 193-204.

Examples

```
data(shrews)
table(c(shrews[,2:3])) # allele counts of locus X9
```

specimen	<i>A small simulated data.</i>
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Description

Format your own data according to this sample dataset for use in AFM.

Usage

```
data(specimen)
```

Format

A data frame with 100 observations on the following 21 variables.

```
subpop  subpopulation code
loc1.1  allelic copy 1, locus 1
loc1.2  allelic copy 2, locus 1
loc2.1  allelic copy 1, locus 2
loc2.2  ...
loc3.1  ...
loc3.2  ...
loc4.1  ...
loc4.2  ...
loc5.1  ...
loc5.2  ...
loc6.1  ...
loc6.2  ...
loc7.1  ...
loc7.2  ...
loc8.1  ...
loc8.2  ...
loc9.1  ...
loc9.2  ...
loc10.1 ...
loc10.2 ...
```


Details

Each row represents one individual. Column 1 contains the subpopulation code (can be any number or string). Columns 2+ contain the genotypic information; genotypes can be coded by any number or string. Each number/string is one haploid genotype, while suites of two columns represent diploid genotypes. Missing data must be coded by NA. (Missing the other half of a diploid genotype is ok.) Subpopulation codes must NOT be missing, and there has to be at least two subpopulations. (Insert a dummy row, if you want to estimate within-population theta for one subpopulation.)

Examples

```
data(specimen)
head(specimen) # how it looks
write.table(specimen, row.names=FALSE, file="specimen.txt") # for reference
```

sticklebacks

Data from Nordic ninespine sticklebacks (Pungitius pungitius).

Description

183 fish from 4 local populations sampled for 12 microsatellite loci.

Usage

```
data(sticklebacks)
```

Format

A data frame with 183 observations on the following 25 variables.

```
subpop  subpopulation code
X1125P.1 allelic copy 1, marker X1125P
X1125P.2 allelic copy 2, marker X1125P
X4174P.1 allelic copy 1, marker X4174P
X4174P.2 allelic copy 2, marker X4174P
X7033P.1 allelic copy 1, marker X7033P
X7033P.2 ...
Stn49.1 ...
Stn49.2 ...
Stn96.1 ...
Stn96.2 ...
Stn100.1 ...
Stn100.2 ...
Stn130.1 ...
Stn130.2 ...
Stn163.1 ...
Stn163.2 ...
```

```
Stn173.1 ...  
Stn173.2 ...  
Stn196.1 ...  
Stn196.2 ...  
Stn198.1 ...  
Stn198.2 ...  
Stn380.1 ...  
Stn380.2 ...
```

Details

Each row represents one individual. In this case, the genotypes are microsatellite fragment lengths. Missing data is coded as NA.

Source

Shikano T, Shimada Y, Herczeg G and Merila J (2010): History vs. habitat type: explaining the genetic structure of European nine-spined stickleback (*Pungitius pungitius*) populations. *Molecular Ecology* 19: 1147-1161.

Examples

```
data(sticklebacks)
```

Index

*Topic **datasets**

shrews, [7](#)

specimen, [8](#)

sticklebacks, [9](#)

AFM, [2](#)

do.all, [3](#)

gen.fst, [4](#)

gen.theta, [5](#)

heat.theta, [6](#)

RAFM (*RAFM-package*), [1](#)

RAFM-package, [1](#)

shrews, [7](#)

specimen, [8](#)

sticklebacks, [9](#)